TEACHER GUIDE

Pearson Edexcel International GCSE in Business (4BS1)

Business calculations

Section 3
3.3 Costs and break-even; 3.4 Financial documents



A revision lesson. Pupils run a car dealership in firms of four and revise business calculations on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Total monthly costs

Round 1

A month at the yard.

Total costs for the month?

Fixed costs, a month	**£8,000**
You pay per car	**£17,500**
Cars sold	**4 cars**

ANSWER

£78,000

£8,000 + 4 × £17,500 = £78,000

Why: Fixed costs stay the same. Variable costs grow with every car.

2 Fixed or variable

Round 1

You list your dealership's costs for a month.

Which two are fixed costs?

- **A** Valeting each car sold
- **B** Rent on the showroom
- **C** Cars bought to sell
- **D** Commission on each sale
- **E** The manager's salary

ANSWER

✓ Rent on the showroom
✓ The manager's salary

Why: Rent and salaries stay the same however many cars you sell.

3 Rank by profit

Round 2

Three months: cars sold, the price, and total costs.

Order by profit, highest first?

- **A** 6 at £15,000, costs £75,000
- **B** 20 at £5,000, costs £90,000
- **C** 10 at £9,000, costs £70,000

ANSWER

1. 10 at £9,000, costs £70,000
2. 6 at £15,000, costs £75,000
3. 20 at £5,000, costs £90,000

Why: Profit: £20,000, £15,000, £10,000. The most revenue made the least profit.

Topic tasks

4 Average unit cost

Round 2

Cost of each car sold.

Average cost per car?

Fixed costs, a month	**£2,500**
Variable cost per car	**£9,500**
Cars sold	**5 cars**

ANSWER

£10,000

£2,500 ÷ 5 + £9,500 = £10,000

Why: Each car carries £500 of the fixed costs.

5 More sales or cuts

Round 3

Each car adds £400 (price minus cost); you sell 15 a month. Sell 5 more, or cut fixed costs £1,500?

Which raises profit more, and why?

A Cut fixed costs by £1,500

B Sell 5 more cars

ANSWER

✔ **Sell 5 more cars**

Why: 5 × £400 = £2,000, more than the £1,500 saved.

6 Profit after rise

Round 3

Costs are going up.

Profit after the rise? Minus allowed.

Sales	**£100,000**
Variable costs	**£45,000**
Fixed costs	**£15,000**
Variable costs rise by	**10%**

ANSWER

£35,500

£100,000 − £45,000 − 10% (£4,500) − £15,000 = £35,500

Why: The rise adds £4,500 to variable costs.

Final questions

Each pupil answers 6 final questions on business calculations. They come from these 11 questions.

Then each pupil writes one short answer: “Explain the difference between fixed and variable costs.”

1 A dealership earns revenue of £86,000 and has total costs of £68,500 in a trading period. What is its profit?
✔ **£17,500**

2 A vehicle retailer has sales revenue of £150,000 and cost of sales of £90,000. What is its gross profit?
✔ **£60,000**

3 A showroom has fixed costs of £11,500 and variable costs of £37,500 for a month. What are total costs?
✔ **£49,000**

4 A showroom expects £72,000 sales revenue and wants to make £23,000 profit in a month. What is the highest total cost consistent with that target?
✔ **£49,000**

5 A used-car retailer earns £95,000 sales revenue and £33,000 gross profit. What was its cost of sales?
✔ **£62,000**

6 A dealer initially expects £70,000 revenue and £66,000 total costs. It then refunds £5,000 of sales while costs stay unchanged. What is the revised result?
✔ **A £1,000 loss**

7 A dealership’s statement of financial position lists an unpaid bank loan. How should the loan be classified?
✔ **As a liability**

8 A campaign leads to the sale of six vehicles at £14,500 each. What is sales revenue?
✔ **£87,000**

9 Revenue is £84,000 and cost of sales £54,000. A supplier discount cuts cost of sales by 10%; revenue is unchanged. What is the new gross profit?
✔ **£35,400**

10 Operations has a total cost budget of £56,000, including £9,500 of fixed handling costs. How much can be spent on variable purchasing costs?
✔ **£46,500**

Final questions

11 Revenue = selling price × quantity sold. A dealer sells 14 vans at £21,000 each. Using the formula, what is its van revenue?

✔ **£294,000**